Logarithmic Functions & Properties

ANSWER KEY

Converting between exponential and logarithmic form

1 Rewrite $2^5 = 32$ in logarithmic form, and $\log_3 81 = 4$ in exponential form.
$2^5 = 32 \Leftrightarrow \log_2 32 = 5$. $\log_3 81 = 4 \Leftrightarrow 3^4 = 81$.

2 Evaluate $\log_5 125$ and $\log_2 \frac{1}{8}$ without a calculator.
$\log_5 125 = 3$ since $5^3 = 125$. $\log_2 \frac{1}{8} = -3$ since $2^{-3} = \frac{1}{8}$.

Properties of logarithms

3 Use logarithm properties to expand $\log\left(\frac{x^3\sqrt{y}}{z^2}\right)$ fully.
$\log\left(\frac{x^3\sqrt{y}}{z^2}\right) = \log(x^3) + \log(y^{1/2}) - \log(z^2) = 3\log x + \frac{1}{2}\log y - 2\log z$.

4 Combine $2\log x - \frac{1}{3}\log y + \log z$ into a single logarithm.
$2\log x - \frac{1}{3}\log y + \log z = \log(x^2) - \log(y^{1/3}) + \log z = \log\left(\frac{x^2 z}{y^{1/3}}\right)$.

5 Use the change of base formula to evaluate $\log_7 50$, to three decimal places.
$\log_7 50 = \frac{\ln 50}{\ln 7} = \frac{3.912}{1.946} \approx 2.010$.

Visualizing logarithmic functions

6 The graph shows $f(x) = \ln(x)$, the natural logarithm.

a State the domain and the vertical asymptote shown in the graph. Domain: $x > 0$. Vertical asymptote at $x = 0$, where the curve plunges toward $-\infty$ as $x \to 0^+$.

b State the x-intercept shown. x-intercept at $(1, 0)$, since $\ln(1) = 0$.

Solving exponential equations using logarithms

7 Solve $5^x = 40$ for x, to three decimal places.
Taking $\log$ of both sides: $x\log 5 = \log 40$, so $x = \frac{\log 40}{\log 5} \approx \frac{1.602}{0.699} \approx 2.292$.

8 Solve $3(2)^{x-1} = 45$ for x, to three decimal places.
$2^{x-1} = 15$. Taking $\log$: $(x-1)\log 2 = \log 15$, so $x - 1 = \frac{\log 15}{\log 2} \approx 3.907$, giving $x \approx 4.907$.

Solving logarithmic equations

9 Solve $\log_2(x+3)=4$ for x, and check that the solution keeps the argument positive.

$x+3=2^4=16$, so $x=13$. Check: $x+3=16>0$ ✓. Solution: $x=13$.

10 Solve $\log(x)+\log(x-3)=1$ for x, checking both solutions against the domain.

Combine: $\log[x(x-3)]=1$, so $x(x-3)=10$: $x^2-3x-10=0$, $(x-5)(x+2)=0$: $x=5$ or $x=-2$. Since $x>0$ and $x-3>0$ are required (domain of the original logs), $x=-2$ is rejected. Solution: $x=5$.

Applications: the pH scale and decibels

11 The pH of a solution is given by pH $=-\log[H^+]$, where $[H^+]$ is the hydrogen ion concentration in mol/L. Find the pH of a solution with $[H^+]=3.2\times10^{-5}$ mol/L, to two decimal places.

pH $=-\log(3.2\times10^{-5})\approx-(-4.495)=4.49$.

12 Sound intensity level in decibels is $L=10\log\left(\frac{I}{I_0}\right)$, where I_0 is a reference intensity. If one sound is 1000 times more intense than another, find the difference in their decibel levels.

Difference $=10\log(1000)-10\log(1)=10\log(1000)=10(3)=30$ decibels.

Transformations of logarithmic functions

13 Describe how the graph of $g(x)=\ln(x+2)-3$ is transformed from the parent function $f(x)=\ln(x)$, and state its new vertical asymptote and domain.

$g(x)=\ln(x+2)-3$ shifts $f(x)=\ln(x)$ left 2 units and down 3 units. The vertical asymptote shifts from $x=0$ to $x=-2$, and the domain becomes $x>-2$.

14 Find the domain of $h(x)=\log(4-x^2)$.

Require $4-x^2>0$: $x^2<4$, so $-2<x<2$. Domain: $(-2,2)$.

Solving equations that combine exponentials and logarithms

15 Solve $e^{2x}-5e^x+6=0$ by substituting $u=e^x$.

Let $u=e^x$: $u^2-5u+6=0$, so $(u-2)(u-3)=0$, giving $u=2$ or $u=3$. Since $u=e^x>0$ for all real x, both are valid: $e^x=2\Rightarrow x=\ln 2$, or $e^x=3\Rightarrow x=\ln 3$.

16 Solve $\log_2(x)+\log_2(x-2)=3$ for x, checking the domain.

Combine: $\log_2[x(x-2)]=3$, so $x(x-2)=2^3=8$: $x^2-2x-8=0$, $(x-4)(x+2)=0$: $x=4$ or $x=-2$. Domain requires $x>0$ and $x-2>0$ (i.e. $x>2$), so $x=-2$ is rejected. Solution: $x=4$.

Comparing logarithmic growth to other function families

17 Explain why logarithmic functions grow much more slowly than linear functions for large x, and give a numerical comparison of $\log(x)$ and x at $x = 1000$.

Logarithms compress large inputs: each tenfold increase in x adds only 1 to $\log(x)$, while a linear function adds proportionally to the increase in x itself. At $x = 1000$: $\log(1000) = 3$, while the linear function $y = x$ gives 1000 — the logarithm is vastly smaller, illustrating its extremely slow long-run growth.

Modeling with logarithms: the Richter scale

18 Earthquake magnitude on the Richter scale is $M = \log\left(\frac{I}{I_0}\right)$, where I_0 is a reference intensity. An earthquake measures magnitude 7, while another measures magnitude 5. Find how many times more intense the magnitude-7 earthquake is.

$7 - 5 = \log\left(\frac{I_7}{I_0}\right) - \log\left(\frac{I_5}{I_0}\right) = \log\left(\frac{I_7}{I_5}\right) = 2$, so $\frac{I_7}{I_5} = 10^2 = 100$. The magnitude-7 earthquake is 100 times more intense.

19 If an earthquake's intensity is $10{,}000$ times a reference intensity I_0, find its Richter magnitude.

$M = \log(10000) = \log(10^4) = 4$.